# *Influence Of Precipitating Agents On Surface Texture And Magnetic Properties Of $Dy_2O_3$ Nanopowder*

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***Abstract*— $Dy_2O_3$ nanopowder was synthesized by co-precipitation method using various precipitating agents like as NaOH, KOH, $NH_4OH$. The prepared samples were characterized for their structural, functional, morphological, optical and magnetic properties by means of XRD, SEM, Micro Raman, PL, FTIR and VSM. XRD patterns revealed that the prepared samples possess cubic structure with lattice planes (222) (400) (440) (622). SEM images displayed agglomerated grains, nanosheet and nanoaggregates for nanoparticles synthesized using different precipitating agents. In Raman spectra, the observed Raman $F_g$ modes were well assigned to cubic $Dy_2O_3$. In FTIR analysis, the peak at ~563cm$^{-1}$ was assigned due to the bending modes of vibrations of Dy-O($\nu_{Dy-O}$) band. The PL spectra of $Dy_2O_3$ have exhibited both yellow and blue emission regions. VSM analysis concluded that the prepared samples are being in paramagnetic nature.**

***Keywords*— *Dysprosium oxide, Rare earth compounds, co-precipitation, paramagnetic property***

## I. INTRODUCTION

Rare earth sesquioxides(REO) showed great scientific and technological interest for many years due to their attractive physical and chemical properties. In these, magnetic nanoparticles have received special attention over the last years[1-2]. These nanoparticles are widely used in high-density magnetic recording, magneto-caloric applications, magnetically guided drug delivery, catalysis, biosensors, biocompatible magnetic nanoparticles, biosensors, and magneto-optical devices[3-6]. Among different types of magnetic materials, Dysprosium oxide($Dy_2O_3$) appeals to researchers due to its cubic magneto-crystalline anisotropy, high resistivity, large energy band and high dielectric constant[7-9]. $Dy^{3+}$ are very important activators for the luminescent materials. $Dy_2O_3$ is used as a photoluminescent and thermoluminescent material. As an activator, $Dy^{3+}$ has been extensively studied in various hosts due to its unique spectral properties[10]. The luminescent properties of these materials make them attractive for many technological applications like display devices, up-conversion solar cells, white-light generation and detectors in medical diagnosis equipment among others[11]. Recently, dysprosium oxide was applied as $T_2$, magnetic resonance imaging(MRI) contrast agents[12]. Over the past years several physical and chemical routes used for the preparation of rare earth sesquioxides namely $Eu_2O_3$, $La_2O_3$, $Dy_2O_3$, $Nd_2O_3$ etc.,[13-15].

Co-precipitation was one of the most promising technique for the preparation of nanopowders on a large scale in water rather than in organic chemicals without using special type of equipments, hence in the present study co-precipitation has been chosen to prepare $Dy_2O_3$ nanopowder. The prepared nanopowder are subjected to be investigated on structural, magnetical, morphological, Raman shift, functional group and luminescent properties of Dysprosium oxide nanopowder.

## II. EXPERIMENTAL DETAILS

In this procedure, 0.057M concentration of Dysprosium nitrate penta hydrate solution was prepared by dissolving in de-ionized water. Then three different precipitation agents are used such as Sodium hydroxide(NaOH), Potassium hydroxide(KOH), Ammonium hydroxide($NH_4OH$) with 3M was added into the above solution. The pH was maintained at 12. The mixture was stirred for 30min using magnetic stirrer. The white precipitate was filtered, washed repeatedly for several times with distilled water and ethanol to residual ammonia and nitrate ions. After washing, the resultant product was dried at $80^0$ C for 10h. A portion of the precipitate was annealed at $250^0$ C and $500^0$ C for 2h.

## III. Results and Discussion

### A. *X-ray Diffraction Analysis*

The XRD pattern of different precipitating agents and various annealed temperature of the $Dy_2O_3$ nanopowder is shown in Fig 1(a),(b),(c). As prepared ($D_1$,$D_4$,&$D_7$) and annealed at $250^0$ C($D_2$,$D_5$&$D_8$) show an amorphous in nature.

When the obtained powders annealed at $500^0$ C using different precipitating agents such as NaOH($D_3$), KOH($D_6$), $NH_4OH$($D_9$) exhibits the plane (222), (400), (440), (622) at 2θ=28.92°, 33.33°, 48.17°, 57.23° respectively which were indexed and good agreement with JCPDS card no. 43-1006 [16]. These emerging planes are due to $Dy_2O_3$ which is an indication of crystallization of $Dy_2O_3$. The most intensity peak observed in all samples at an angle 2θ=28.92° which corresponds to (222) reflection plane with full width at half maximum of 0.29° which represents that the product is highly crystalline [17].

The average particle size, dislocation density, and micro strain were calculated for the reflections using Debye Scherer's formula. The calculated data has been reported in table 1. From the close observation of table 1, the sample prepared using NaOH as precipitation agent exhibits a better crystalline that that of other precipitating agents. As crystalline size increased, dislocation density and micro strain are decreased which reveals that the sample prepared using NaOH as precipitation agent shows good crystalline and less defects.

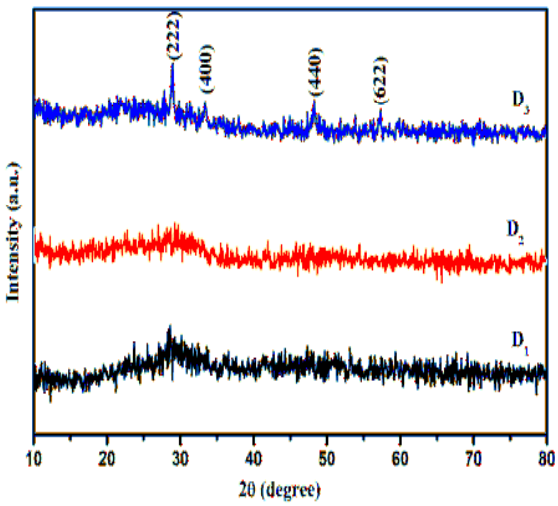



**Fig 1(a).** XRD patterns of $Dy_2O_3$ nanoparticle using NaOH as a precipitation agent: ($D_1$) as-prepared, ($D_2$) annealed at $250^0$ C and ($D_3$) annealed at 500°C.

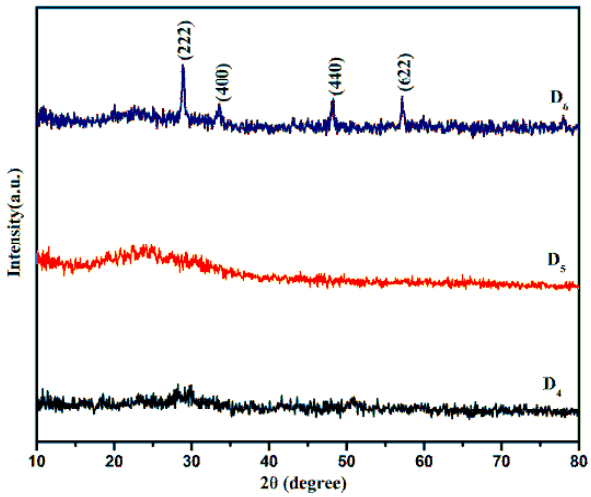



**Fig 1(b)**. XRD patterns of $Dy_2O_3$ nanoparticle using KOH as a precipitation agent: ($D_4$) as-prepared, ($D_5$) annealed at 250°C and ($D_6$) annealed at 500°C

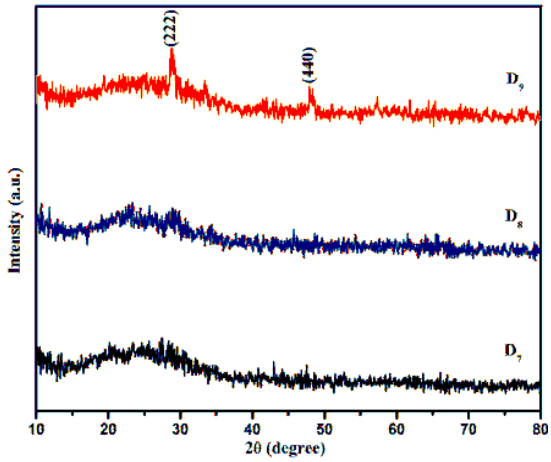



**Fig. 1 (c)** XRD patterns of $Dy_2O_3$ nanoparticle using $NH_4OH$ as a precipitation agent: ($D_7$) as-prepared, ($D_8$) annealed at 250°C, ($D_9$) annealed at 500°C.

**Table. 1** shows the crystalline size, Dislocation density and Micro strain for as prepared nanopowder

| Samples | Crystalline size D(nm) | Dislocation density $\times 10^{14}$ (lines/m$^2$) | Strain $\times 10^3$(lines$^{-2}$.m$^{-4}$) |
|---|---|---|---|
| $D_3$ | 72.73 | 18.90 | 4.76 |
| $D_6$ | 48.48 | 42.54 | 7.14 |
| $D_9$ | 48.46 | 42.58 | 7.18 |

### *B. Raman Spectroscopic Analysis*

Raman active phonon spectra of various precipitating agents annealed temperature of $Dy_2O_3$ nano powders structure again confirmed by the XRD analysis report (Fig 2). This confirmation is good agreement with Chandrasedhar and Nagabhushana and Bahaa et.al.,[18]. According to group theory, 22 Raman active modes were predicted for C-type $Dy_2O_3$ structure were also observed. The observed lines at 370 $cm^{-1}$ which was mainly assigned to the $F_g$ modes and combination of $A_g$+$E_g$ modes [19]. The observed Raman modes are well assigned to cubic $Dy_2O_3$.

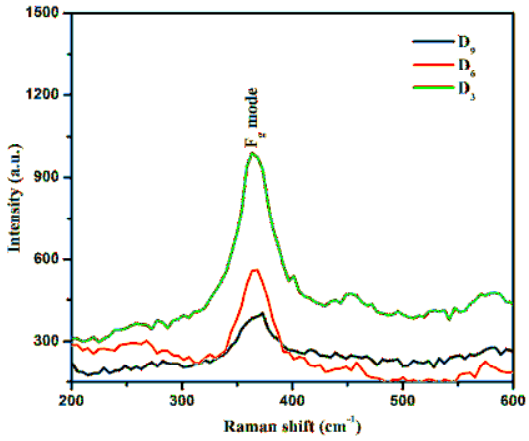



**Fig. 2** Raman spectra of $Dy_2O_3$ nanoparticle using different precipitant agents annealed at 500℃:NaOH($D_3$), KOH($D_6$), $NH_4OH$($D_9$)

### *C.FTIR Analysis*

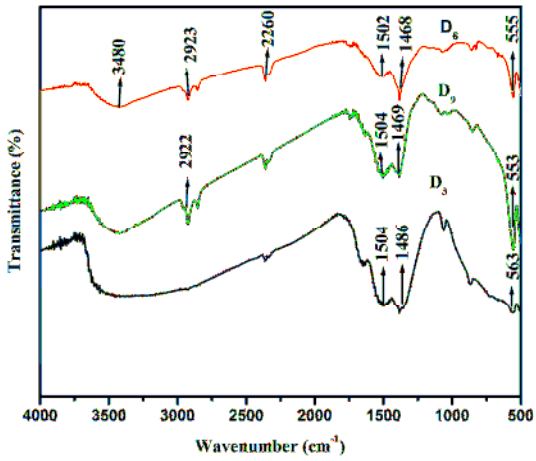



**Fig. 3** FTIR spectra of prepared nanopowder at various precipitating agent and annealed at 500°C.

Fig 3 shows the Fourier Transform Infrared spectra of $Dy_2O_3$ nanopowders. The broad absorption band around 3480 $cm^{-1}$ in all the samples were assigned to the stretching vibration of O-H bond. A small peak centered at 2923 $cm^{-1}$ were attributedthe asymmetric stretching vibration of C-H bond. The peak at 2260 $cm^{-1}$ has been assigned to the C-N stretching vibration. The absorption bands around at 1504 $cm^{-1}$ and 1469 $cm^{-1}$ are ascribed to the $\nu_3$ mode of $CO_3^{2-}$ and C-O stretching vibration respectively. The peak at ~563 $cm^{-1}$ was assigned to the bending vibration of Dy-O($\nu_{Dy-O}$) bond [20-21].

### *D. Photoluminescence Studies*

The optimized samples PL spectra of $Dy_2O_3$ nanopowder prepared using different precipitant agents are shown in Fig 4. The emission spectra of the samples are having peaks at 481 and 521 nm respectively. The peaks are mainly attributed to the transition of $Dy^{3+}$ ions, $^4F_{9/2} \rightarrow {}^6H_{15/2}$ and $^4F_{9/2} \rightarrow {}^6H_{13/2}$ respectively[22]. $Dy^{3+}$ ion has been proved to be a potential candidates for the development of yellow laser as $Dy^{3+}$ ion exhibits two bands in the blue (470-500 nm) and yellow (570-600 nm) regions [23]. When $Dy^{3+}$ ion is located at low symmetry local site (without inversion symmetry), sthe yellow emission is often dominant in the emission spectrum, when $Dy^{3+}$ is presented at a high symmetry local site (with inversion symmetry). The blue emission is stronger than the yellow emission and it dominates in the emission spectrum [24-25]. So it is well known that $Dy^{3+}$ emission at 481 nm has been due to magnetic dipole, and the emission at 521 nm is owing to an electric dipole. The 521 nm is predominant only when $Dy^{3+}$ ions are located at low symmetry sites with no inversion center.

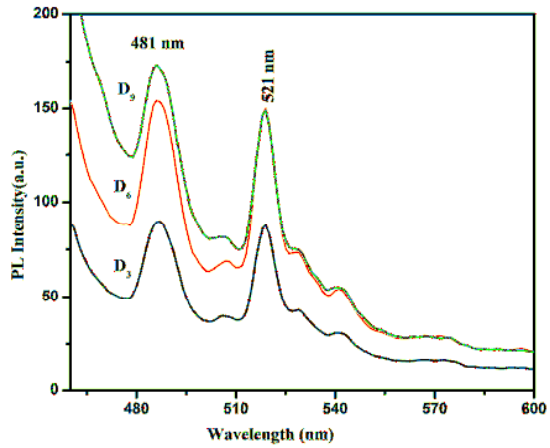


Fig 4 PL spectra of $Dy_2O_3$ nanopowder using NaOH($D_3$), KOH($D_6$), $NH_4OH$($D_9$) different precipitant agents.

*E. Magnetic properties*

Magnetic properties of the optimized $Dy_2O_3$ nanoparticles are studied using vibrating sample magnetomenter in the fieled range between 2 to -2 testla at room temperature shown in Fig (5). In this fig no saturation of the magnetic momentum is observed even upto two tesla and all the samples have zero correctivity and remnant magnetization that means no hysteresis loop [26]. Hence one can conclude the prepared samles are being in paramagnetic nature. The increasing magnitude of magnetization with respect to applied magnetic field of the optimized samples may be due to increasing concentration of $Dy^{3+}$ ions, or oxygen vacancies present in the basic compounds which is also confirmed by photoluminescence analysis.

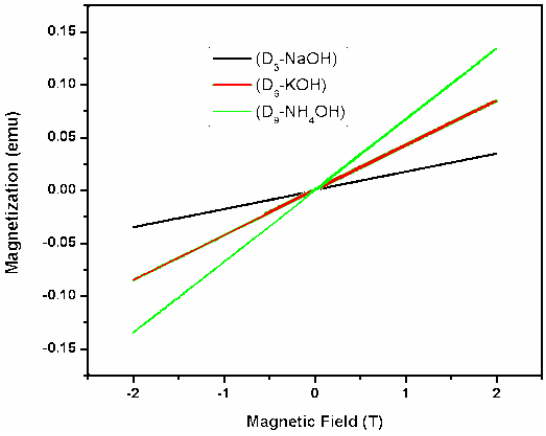
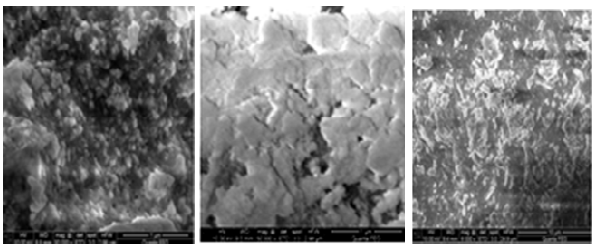



**Fig. 5** Magnetometric curves for $Dy_2O_3$ nanopowder at room temperature

*F. **Morphological properties***

The $Dy_2O_3$ nanopowder were prepared by co-precipitation method annealed at $500^0$ C using different precipitation agents. Fig 6(a) shows typical Scanning Electron Micrograph (SEM) of $Dy_2O_3$ nanopowder prepared by using (NaOH) as a precipitation agent, it shows that the agglomerated grains are found. The size of grains is varied in range of 168-420 nm. Fig 6(b) shows an SEM image of the $Dy_2O_3$ nanopowder prepared using KOH as a precipitation agent, it could be seen that the particle size of $Dy_2O_3$ nanopowder increases to 392-560 nm. Fig 6 (c) shows an SEM image of the nanopowder using precipitating agent as $NH_4OH$ , which reveals that the formed $Dy_2O_3$ nanopowder consist of aggregated grains , and the size of the grains is varied between 625 nm-2500 nm, and the surface of the grain is rough [27-28].



*(a)* *(b)* *(c)*

**Fig. 6** SEM image of $Dy_2O_3$ nanopowder annealed at 500°C using different precipitating agents (a) NaOH, (b) KOH and (c) $NH_4OH$.

## IV. Conclusion

$Dy_2O_3$ nanopowders were synthesized by co-precipitation method using various precipitation agents like as NaOH, KOH and $NH_4OH$ successfully. XRD patterns revealed that the prepared nanopowder annealed at 500°C exhibited cubic structure. Raman active modes of $Dy_2O_3$ nanopowder revealed on obviously C-type of Dysprosium oxide which are confirmed by X-ray diffraction analysis. The bending vibration of $Dy_2O_3$ at 563 $cm^{-1}$ was detected by Fourier Transform Infrared Spectroscopy. Emission spectra of all the samples shows peaks at 481 and 521 nm which were attributed to the transition of $Dy^{3+}$ ions. VSM analysis conclude the prepared samples are being in paramagnetic nature. Finally, it is concluded that the structural, optical and magnetic property of the synthesized $Dy_2O_3$ nanopowders depend upon the precipitating agent using in the co-precipitation method. Among various precipitation agents, NaOH gives well defined highly crystalline nature.

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